

# Chapter 1 : Introduction

20-10-2020

Matter

- solid
- liquid
- gas

- atoms

- molecules:  $H_2O$ ,  $CH_4$ ,  $NO_2$

\* distance between atoms

\* intermolecular forces.

- element  
compound

like dissolves like

properties

- Chemical
- physical

- Mixture

- Homogenous
- Heterogenous

- units of measurement  $\Rightarrow$  SI unit (standard international unit)

time: S      mass: kg

Temperature

- $^{\circ}C$
- $^{\circ}F$
- K (SI unit)

Volume

- $m^3$
- L
- mL
- $cm^3$

$$0^{\circ}C = 273 K$$

$$^{\circ}C = \frac{5}{9} (^{\circ}F - 32)$$

$$1 m^3 = 10^6 cm^3$$

\* Convert 460  $^{\circ}F \rightarrow K$

$$1 m^3 = 10^3 L$$

$$^{\circ}C = \frac{5}{9} (460 - 32)$$

$$1 L = 10^3 mL = 10^3 cm^3$$

$$= \frac{5}{9} \times 428$$

$$cm^3 = mL$$

$$= 237.7 \rightarrow ^{\circ}C + 273 = K$$

$$K = 510.7$$



# Dimensional Analysis

$$d = \frac{\text{mass}}{\text{volume}}$$

density

\* Convert 12.5 g/cm<sup>3</sup> into Kg/L

$$12.5 \frac{\text{g}}{\text{cm}^3} \times \frac{10^3 \text{ cm}^3}{1 \text{ L}} \times \frac{1 \text{ Kg}}{10^3 \text{ g}} = 12.5 \text{ Kg/L}$$

\* Convert 1.5 g/L → ng/m<sup>3</sup>

$$1.5 \frac{\text{g}}{\text{L}} \times \frac{10^9 \text{ ng}}{1 \text{ g}} \times \frac{1 \text{ L}}{10^{-3} \text{ m}^3} = 1.5 \times 10^{12} \text{ ng/m}^3$$

# Significant Figures

$$1.78 \text{ m} \rightarrow 178 \text{ cm} \pm 1$$

3 sig. fig. \* The error is in the third sig. fig.

$$178.6 \pm 0.1$$

4 sig. fig. \* The error is in the 4<sup>th</sup> sig. fig.

$$1.03 \times 10^4$$

3 sig. fig.

\* adding a power does not affect the number of significant figures.

$$0.0025$$

2 significant figures

\* لانا بدأ الرقم بصفر، نبدأ

بالعدد من أول رقم بعد الأصفار

in multiplication & division only

$$3.2 \times 2.478 = 7.9296$$

2 sig. fig. 4 sig. fig. = **7.9**

\* The answer must contain the lowest number of significant figures.

$$4.65 + 3.171 = 7.821$$

2 decimals 3 decimals = **7.82**

\* In addition & subtraction, the answer must contain the lowest number of decimals.

H.W

$$(10.75 - 3.153) = 7.597 = 7.60 = 3.04 = \boxed{3.0}$$

2.5 2.5 2.5



# Dimensional Analysis

$$d = \frac{\text{mass}}{\text{volume}}$$

density

\* Convert  $12.5 \text{ g/cm}^3$  into  $\text{Kg/L}$

$1 \text{ g} = 10^3 \text{ mg}$   
 $= 10^6 \text{ Mg}$   
 $= 10^9 \text{ ng}$   
 $= 10^{12} \text{ pg}$   
 $= 10^{15} \text{ fg}$

$$12.5 \frac{\text{g}}{\text{cm}^3} \times \frac{10^3 \text{ cm}^3}{1 \text{ L}} \times \frac{1 \text{ Kg}}{10^3 \text{ g}} = 12.5 \text{ Kg/L}$$

\* Convert  $1.5 \text{ g/L} \rightarrow \text{ng/m}^3$

$$1.5 \frac{\text{g}}{\text{L}} \times \frac{10^9 \text{ ng}}{\text{g}} \times \frac{\text{L}}{10^{-3} \text{ m}^3} = 1.5 \times 10^{12} \text{ ng/m}^3$$

# Significant Figures

$$1.78 \text{ m} \rightarrow 178 \text{ cm} \pm 1$$

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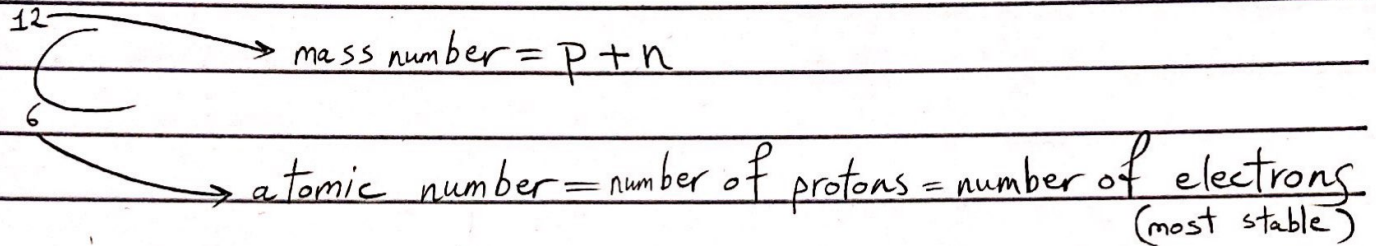
= **3.0**



Precision: How well measurements agree with each other.

Accuracy: How well measurements agree with true value.

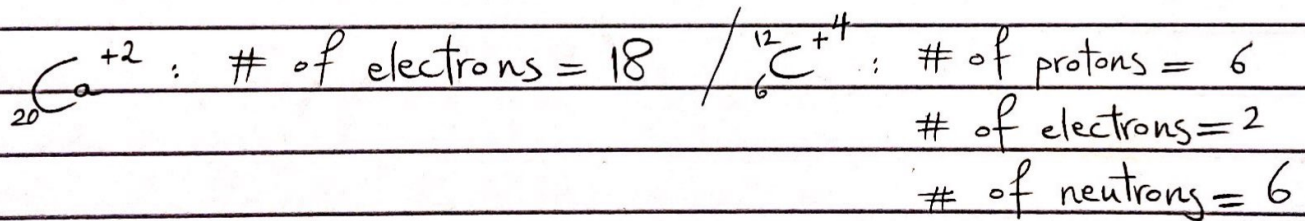
## Chapter 2: Atoms, molecules, & Ions



radioactivity:  $\alpha$ -particles: + charge

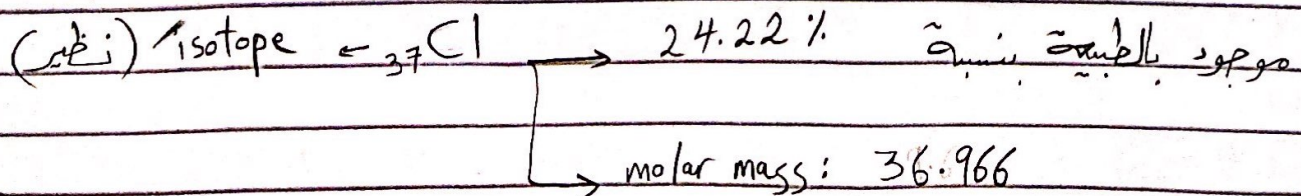
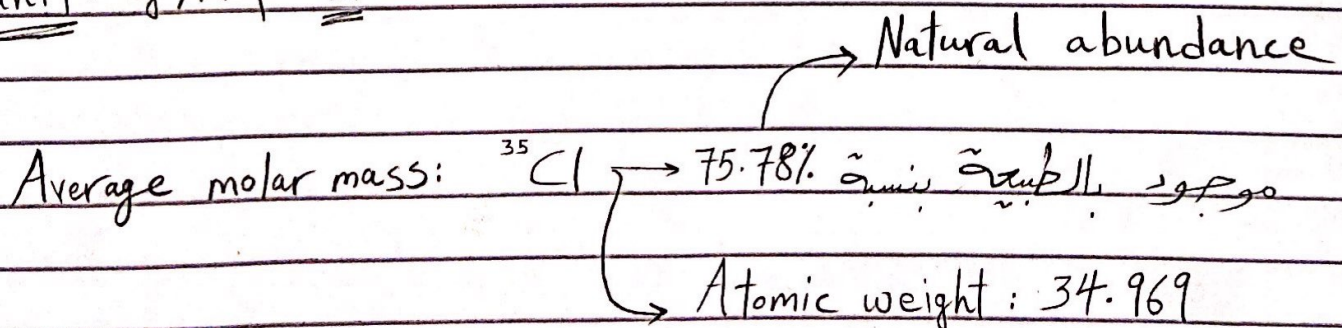
$\beta$ -particles: - charge

$\gamma$ -radiation: high energy radiation.



\* Atomic Weight: (Molar/Molecular mass)

unit: g/mol or amu





Average molar mass for Cl =  $(0.7578 \times 34.969) + (0.2422 \times 36.966)$   
= 35.45 amu

\* Not all elements have isotopes

## Periodic table

Periodic table

Valence

halogens

Period

Group

Transition elements

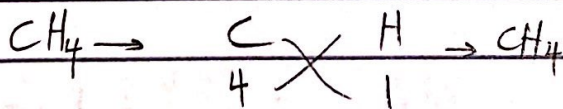
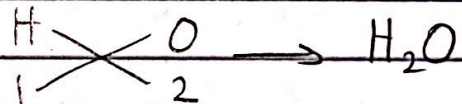
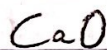
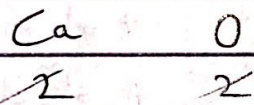
alkali metals

alkali earth metals

Chalogens

Noble gases

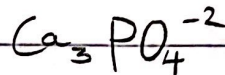
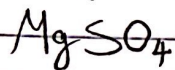
|  |    |    |  |    |    |   |   |    |    |  |
|--|----|----|--|----|----|---|---|----|----|--|
|  | 1  |    |  |    |    |   |   |    |    |  |
|  | H  | 2  |  | 3  | 4  | 3 | 2 | 1  | He |  |
|  | Li | Be |  | B  | C  | N | O | F  | Ne |  |
|  | Na | Mg |  | Al | Si | P | S | Cl | Ar |  |
|  | K  | Ca |  |    |    |   |   |    |    |  |



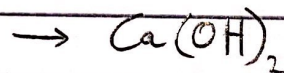
\* When we mention transition elements, we must mention their valence, because they don't have a constant valence.

Ex:  $\text{Fe(II)}$  ,  $\text{Fe(III)}$

\* The charge of complex ions & transition metals is the valence.



\* The valence of all other elements is the valence of its group.



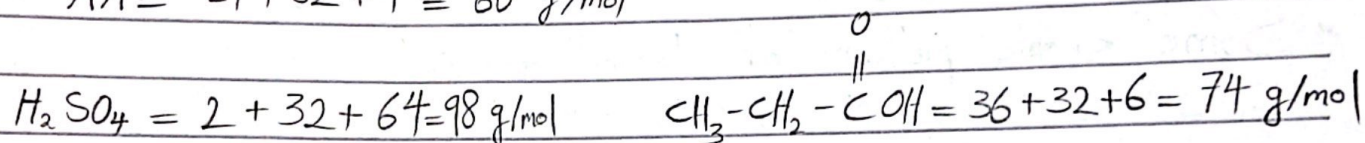


### Chapter 3: Stoichiometry

$$n = \frac{\text{Mass} \rightarrow \text{g}}{\text{Molar Mass} \rightarrow \text{g/mol}}$$

Ex:  $\text{CH}_3\text{COOH}$  C:12, O:16, H:1

$$\text{MM} = 24 + 32 + 4 = 60 \text{ g/mol}$$



Ex: How many moles are there in 20 g NaOH? Na:23, O:16, H:1

$$n = \frac{20}{40} = 0.5 \text{ mol NaOH}$$

Ex: How many milligrams are there in 2.5 mol of  $\text{Ca}(\text{OH})_2$ ?

$$2.5 = \frac{\text{mass}}{74}$$

Ca: 40

O: 16

H: 1

$$\text{mass} = 2.5 \times 74 = 185 \text{ gram} \times \frac{10^3 \text{ mg}}{1 \text{ g}} = 185 \times 10^3 \text{ mg}$$

\* 1 mol  $\rightarrow 6.02 \times 10^{23}$  molecules or atoms

Ex: How many molecules are there in 7 mol/s?

1 mol  $\rightarrow 6.02 \times 10^{23}$  molecules

7 mol  $\rightarrow x$

$$x = 7 \times 6.02 \times 10^{23} \rightarrow x = 42.14 \times 10^{23} \text{ molecules}$$

Q: How many molecules are there in 6 g of NaOH?  
N: 23, O: 16, H: 1

$$n = \frac{6}{23+16+1} = \frac{6}{40} = 0.15 \text{ moles NaOH}$$

1 mol  $\rightarrow 6.02 \times 10^{23}$  molecules

0.15 mol  $\rightarrow x$

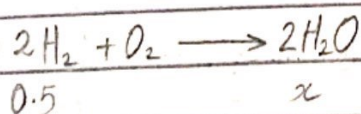
$$x = 0.15 \times 6.02 \times 10^{23} = 0.903 \times 10^{23} \text{ molecules}$$

\* Some simple patterns of chemical reactivity

Reactions  $\left\{ \begin{array}{l} \rightarrow \text{Decomposition: } 2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2 \\ \rightarrow \text{Combination: } 2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO} \end{array} \right.$

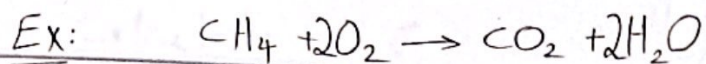
- Balanced equations  $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$

\* How many moles of  $\text{H}_2\text{O}$  when 0.5 mol of  $\text{H}_2$  are reacted?



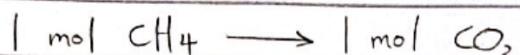
$$2x = 1 \rightarrow x = 0.5 \text{ moles H}_2\text{O}$$





How many grams of  $\text{CO}_2$  were produced when 4g of methane were used

$$4\text{g} \rightarrow \text{mol} : \frac{4}{16} = \frac{1}{4} = 0.25 \text{ mol CH}_4$$



$$0.25 \rightarrow x$$

$$x = 0.25 \text{ mol CO}_2$$

$$0.25 \text{ mol CO}_2 = \frac{\text{mass}}{44}$$

$$\text{mass} = 11 \text{ g CO}_2$$

Empirical Formula:

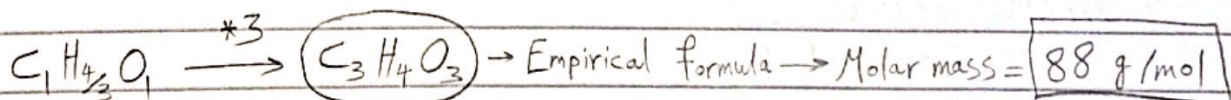
vitamin C  $\rightarrow$  40.92% C  $\text{C}_x\text{H}_y\text{O}_z = ?$  \* نعتبر النسبة المئوية في الـ (Mass)، ثم نستخرج عدد المولات.  
4.58% H  $\text{MM} = 176 \text{ g/mol}$   
54.5% O

$$n_c = \frac{40.92}{12} = 3.41 \text{ mol C}, n_H = \frac{4.58}{1} = 4.58 \text{ mol H}$$

$$n_O = \frac{54.5}{16} = 3.41 \text{ mol O}$$

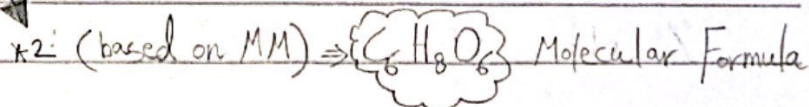
بعد حساب عدد المولات، نقسمهم جميعاً على أقل عدد صحيح

$$\frac{3.41}{3.41} = 1 \text{ C}, \frac{4.58}{3.41} = 1.3 = \frac{4}{3} \text{ H}, 1 \text{ O}$$



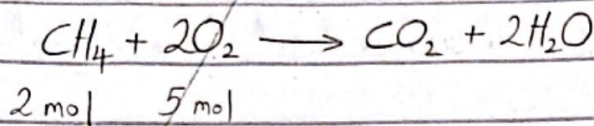
We multiply by 2 to get the given MM

$$88 \times 2 = 176 \text{ g/mol}$$





## Limiting Reactant



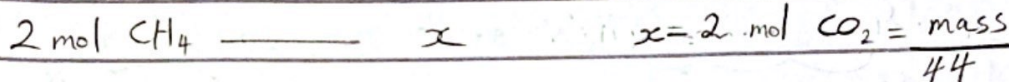
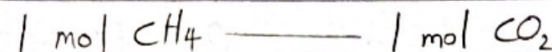
$$\frac{2}{1} = 2$$

أيهما أقل؟

\* نقسم عدد المولات على ال Coefficient

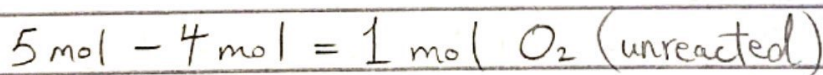
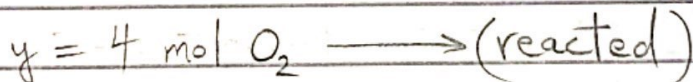
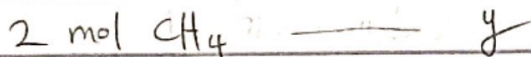
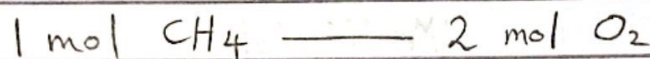
$$\frac{5}{2} = 2.5$$

\* نستخدم المادة التي تُستهلك أولاً...



$$= 88 \text{ g CO}_2$$

\* How many grams of  $\text{O}_2$  were left unreactant?



$$1 \text{ mol} = \frac{\text{Mass}}{\text{Molar mass}}$$

$$\longrightarrow 1 \text{ mol} = \frac{\text{mass}}{32}$$

$$\longrightarrow \text{mass} = 32 \text{ g}$$

$$\% \text{ Yield} = \frac{\text{actual yield}}{\text{Theoretical Yield}} \times 100$$

مثال: إذا أجرينا التجربة السابقة علينا وطلع لدينا 68g

$$\% \text{ Yield} = \frac{65}{88} \times 100 = 73.9\%$$

تمثل نسبة النجاح في عمل التجربة (عادةً تُعتمد 90% فما فوق)

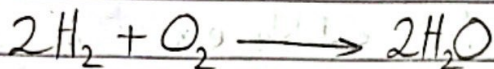


\* Calculate %C in  $C_6H_{12}O_6$

$$\%C = \frac{6 \times 12}{(6 \times 12) + (1 \times 12) + (6 \times 16)} \times 100 = \boxed{40\%}$$

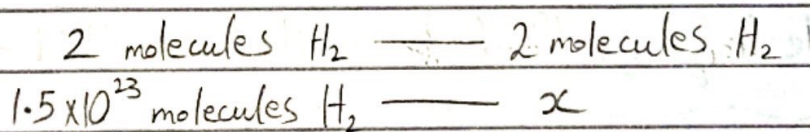
$$\%H = \frac{1 \times 12}{180} \times 100 = \boxed{6.67\%}$$

$$\%O = 100 - 46.67 = \boxed{53.33\%}$$



\* How many molecules of  $H_2O$  are produced if  $1.5 \times 10^{23}$  molecules of  $H_2$  reacted?

\* يمكن قراءة التفاعل بالذرات كما يمكن قراءته بالجولات ، يعني



$$2x = 2 \times 1.5 \times 10^{23} \longrightarrow x = 1.5 \times 10^{23} \text{ molecules of } H_2O$$

\* هاهي الطريقة يمكن اتباعها فقط إذا السؤال أعطانا molecules و طلب molecules .



## Chapter 4: Stoichiometry (II)

### \* General properties of aqueous solutions

- Electrolyte ( $\text{Ca}^{+2}$ ,  $\text{Mg}^{+2}$ , ...)  $\rightarrow$  Conduct electricity, & exist in ionic form.
  - $\downarrow$   
cations

cations

$(\text{Cl}^-, \text{S}^{2-} \dots) \rightarrow$  anions

Non electrolyte:  $C_6H_{12}O_6 \rightarrow$  Water soluble, exist in molecular form.

\* Both electrolytes & Non electrolytes are water soluble.

- Weak electrolytes:  $< 100\%$  decomposition, ex:  $\text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}^+$

- Strong electrolytes: 100% decomposition, ex:  $\text{HCl} \rightarrow \text{H}^+ + \text{Cl}^-$

## # Precipitation Reactions

